Session 19

Adding New Modules in Apache Web Server 2.4

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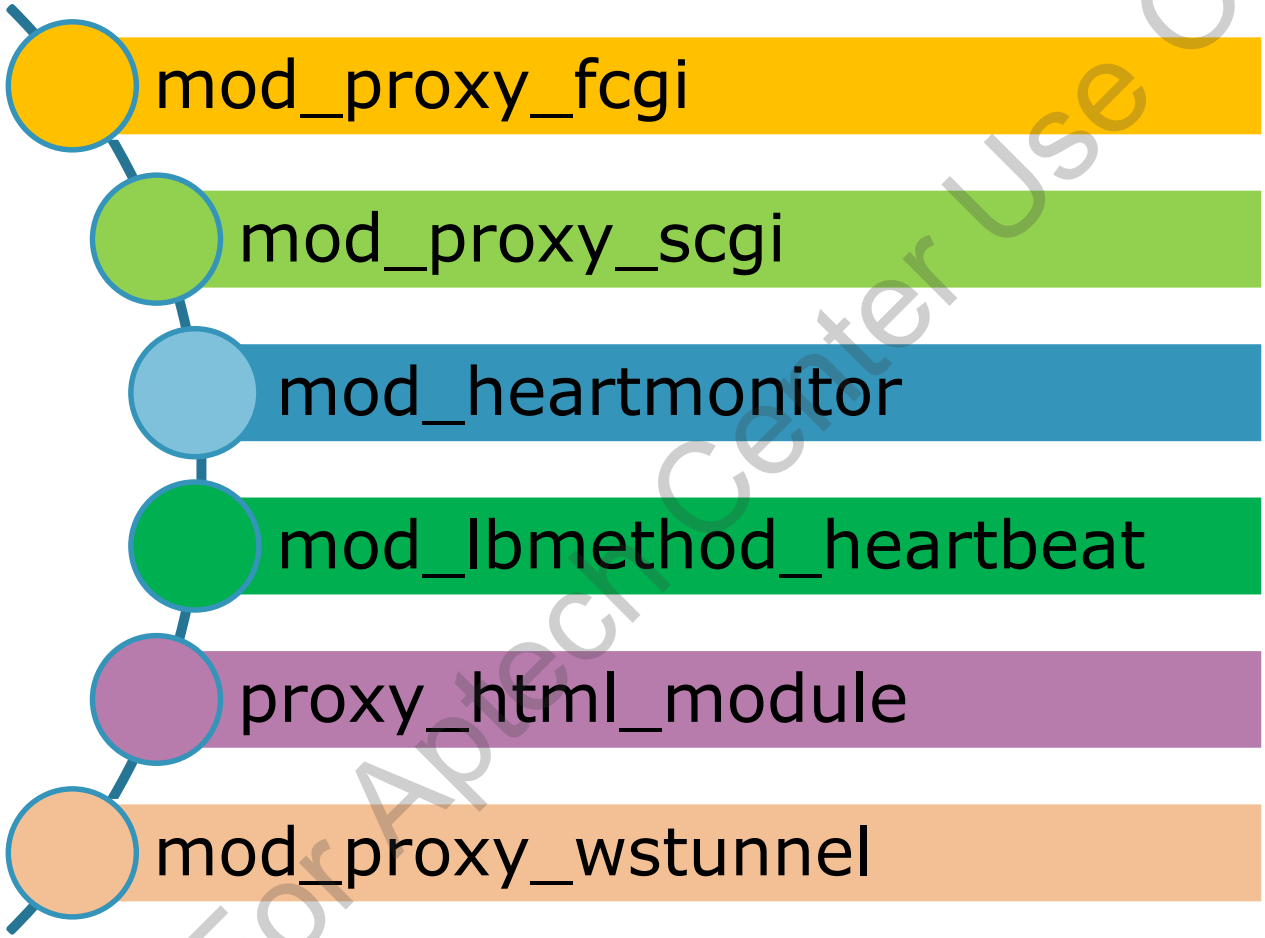
Objectives

- Explain new modules introduced in Apache Web Server 2.4.
- Describe how to configure Apache Web Server 2.4 to support gateway protocols.
- Explain modules related to the multi-protocol proxy/gateway server.
- Explain authentication and authorization modules.
- Describe request processing modules.

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Introducing Modules of the Multi-protocol Proxy/Gateway Server



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Providing Support to FastCGI - mod_proxy_fcgi

- Helps to ease the interaction with CGI programs and Web server.
- By default, the connection reuse functionality is disabled in the `mod_proxy_fcgi` module.
- Enable the connection reuse functionality of a FastCGI application to process concurrent connections:

```
ProxyPass "/aptechapp/"
"fcgi://localhost:4000/" enablereuse=on
```

- Syntax to use the `ProxyPassMatch` directive:

```
ProxyPassMatch [expression] !|url
[key=value [key=value ...]]
```

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Providing Support to SCGI - mod_proxy_scgi

- Is a substitution of the CGI protocol to interact with the Web server.
- Easy to implement as compared to FastCGI.
- Provides the SCGI protocol backend services to `mod_proxy`.
- Requires the `mod_proxy` and `mod_proxy_scgi` services should be available on the server to handle SCGI.

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Storing Server Status Requests - mod_heartmonitor

- An experimental module that keeps record of the messages received from the servers that are enabled with `mod_heartbeat`.

- Provides the server status to `mod_lbmethod_heartbeat` and broadcasts the number of existing connections of the servers.

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Directives of mod_heartmonitor 1-2

HeartbeatListen

- Provides the address of the server that will store the outcome of the servers enabled with `mod_heartbeat`. By default, this directive is disabled.
- Syntax to use the `HeartbeatListen` directive:
  `HearbeatListen addr:port`

HeatbeatMaxServers

- Provides the total number of servers that can send heartbeat request to the server enabled with `mod_heartmonitor`. By default, it can take request of 10 servers.
- Syntax to use the `HeatbeatMaxServers` directive:
  `HeartbeatMaxServers number-of-servers`

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Directives of mod_heartmonitor 2-2

`HeartbeatStorage`

- Provides a storage location to store the data received from the servers enabled with `mod_heartbeat`.
- Syntax to use the `HeartbeatStorage` directive:
  `HeartbeatStorage file-path`

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Distributing Workloads among the Active Connections - mod_lbmethod_heartbeat

- Receives data from `mod_heartmonitor` containing the connection and status information of the servers enabled with `mod_heartbeat`.
- Uses the `HeartbeatStorage` directive to read data of the servers.
- Provides the load balancing schedular algorithm by using `mod_proxy_balancer` to load balance the number of active connections on the servers enabled with `mod_heartbeat`.
- **Syntax**:

```
HeartbeatStorage file-path
```

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Providing Reverse Proxy - proxy_html_module

- Provides the reverse proxy where an URL is rewritten to verify the URL is accessible outside the proxy as well as available publicly.

- Provides the similar functionality that the `ProxyPassReverse` directive provides for the reverse proxy.

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Directives of mod_proxy_html

ProxyHTMLBufSize

Specifies the storage space to store scripts or stylesheets in a buffer

Syntax:
`ProxyHTMLBufSize` bytes

ProxyHTMLCharsetOut

Indicates an encoding for the output of the mod_proxy_html module

Syntax:
`ProxyHTMLCharsetOut Charset | *`

ProxyHTMLDocType

Specifies the document type declaration of XHTML or HTML

Syntax:
`ProxyHTMLDocType XHTML|HTML[Legacy]`

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Directives of mod_proxy_html

ProxyHTMLEnable

Activates or deactivates the proxy_html filter

Syntax:

```
ProxyHTMLEnable
On|Off
```

ProxyHTMLEvents

Denotes the scripting events that triggers proxy

Syntax:

```
ProxyHTMLEvents
attribute
[attribute ...]
```

ProxyHTMLExtended

Sets the filter to fix URLs in scripting events, stylesheets, or inline scripts

Syntax:

```
ProxyHTMLExtended
On|Off
```

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Directives of mod_proxy_html

ProxyHTMLFixups

Resolves the mistakes or errors in HTML code

Syntax:

```
ProxyHTMLFixups
[lowercase]
[dospath] [reset]
```

ProxyHTMLInterp

Activates/deactivates interpolation of map instructions on each request basis

Syntax:

```
ProxyHTMLInterp
On|Off
```

ProxyHTMLLinks

Provides a set of attributes of URL that need to be rewritten for specified element

Syntax:

```
ProxyHTMLLinks
element
attribute
[attribute2 ...]
```

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Directives of mod_proxy_html

ProxyHTMLMeta

Activates or deactivates the metadata section of the HTML code

Syntax:

```
ProxyHTMLMeta
On|Off
```

ProxyHTMLStripComments

Activates or deactivates the scripts mentioned in the HTML comments

Syntax:

```
ProxyHTMLStripCo
mments On|Off
```

ProxyHTMLURLMap

Specifies a rule of rewriting the URLs

Syntax:

```
ProxyHTMLURLMap
from-pattern to-
pattern [flags]
[cond]
```

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Supporting Web-sockets Tunnels - mod_proxy_wstunnel

- Supports tunneling of Web socket connections with `mod_proxy`.
- Uses the `ProxyPass` directive to support the proxying requests to Web sockets server.
- Example of using `mod_proxy_balancer` for load balancing several backend Web sockets server:

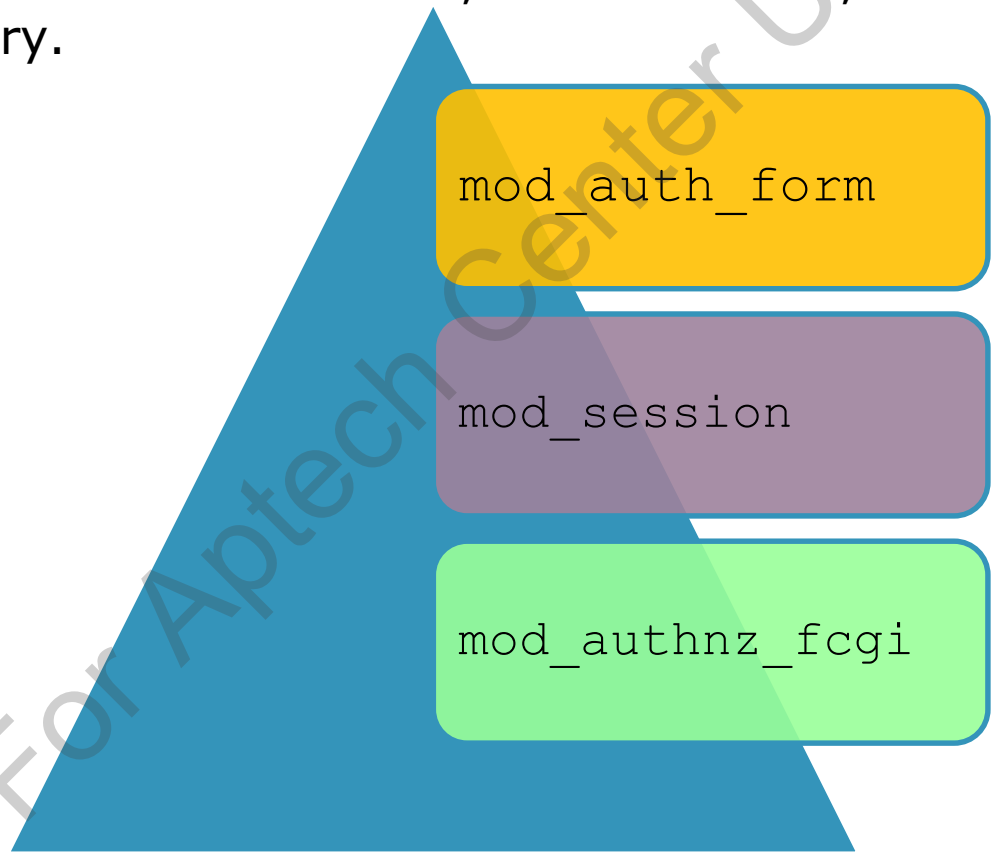
```
ProxyPass "/ws2/"
"ws://websocketexample.com/"
ProxyPass "/wss2/"
"wss://websocketexample.com/"
```

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Introducing Authentication and Authorization Modules

- Apache Web Server 2.4 introduces enhancements in authentication and authorization modules with a new structured code in the access control, authentication, and authorization directory.



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Authenticating User Credentials – mod_auth_form 1-2

- Incorporate HTML login forms to provide a responsive capability to users.
- Helps to restrict users by using the HTTP basic and HTTP digest authentication methods.
- Works with the help of authentication and authorization modules, such as `mod_authn_file` and `mod_authz_user`.



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Authenticating User Credentials – mod_auth_form 2-2

- Example to authenticate a form by specifying the location of the session and the authentication method to be used.

```
AuthFormProvider file
AuthUserFile "conf/passwd"
AuthType form
AuthName realm
AuthFormLoginRequiredLocation
"http://auth_example.com/login_form.html"
Session On
SessionCookieName session path=/
SessionCryptoPassphrase secret
```

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Authenticating User Credentials – mod_request

- Keeps track of the HTTP requests to authenticate user credentials and supports the `KeptBodySize` directive to retain the body of the HTTP request body rather than restricted to the specified size limit.

- Syntax to use the `KeptBodySize` directive:

```
KeptBodySize maximum_size_in_bytes
```

- Example:

```
<Directory
"/var/www/mod_request_test">
   AllowOverride None
   Require all granted
   KeptBodySize 2048
</Directory>
```

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Supporting Sessions – mod_session

- Provides a session that contains the user's credentials if the user is valid.
- Provides support for the sessions that can be created on the server or on the browser.
- Uses the `LoadModule` directive to load the required modules, such as `mod_session_cookie`, `mod_session_crypto`, and `mod_session_dbd` either at run-time or at compile time.

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Example: Create a Browser-based Session

Create a session and specify the cookie name to store the session details.

- Session On
- SessionCookieName cookie_session path=/;domain=simon.test;httponly;

Store information into the session.

- SessionHeader X-Replace-Session

Store information into the session by creating a PHP file and inserting the CGI script.

- #!/bin/bash
- echo "Content-Type: text/plain"
- echo "X-Replace-Session: key1=Aptech&key2=training&key3=Apache"
- echo
- env

Execute the PHP file.

- The PHP file reads the session by viewing the value of the HTTP_SESSION variable

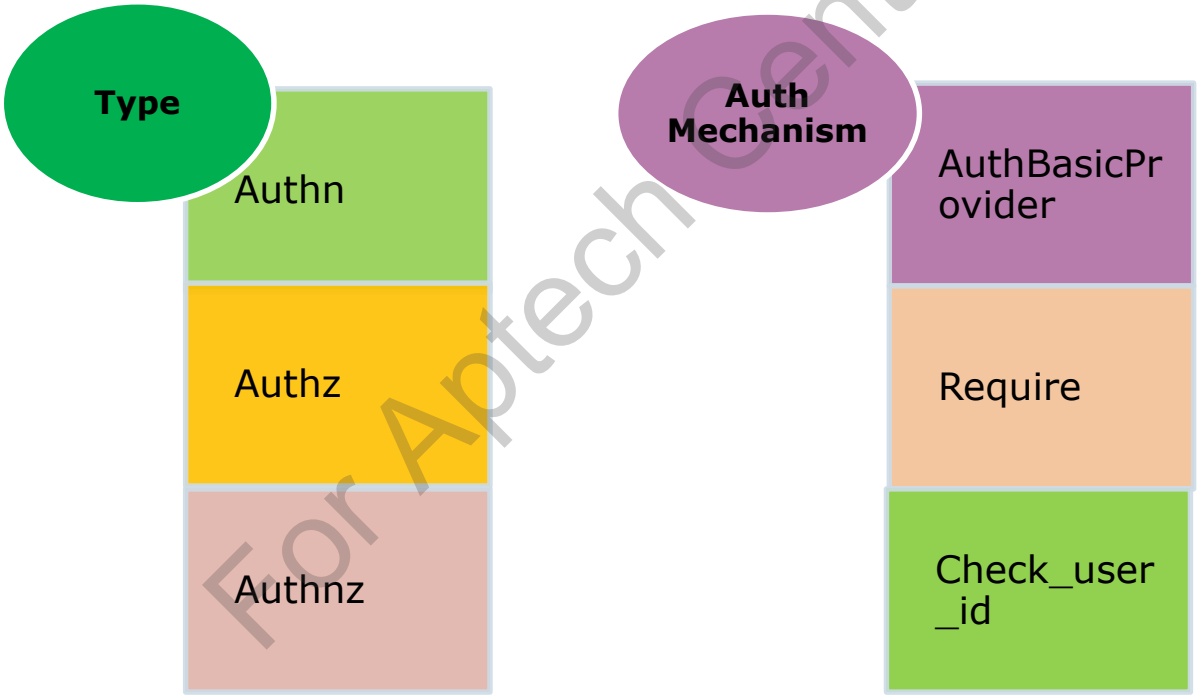
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Validating Users and Approving Resources - mod_authnz_fcgi

- Helps FastCGI authorizer applications to validate users using the Basic authentication method and to approve Apache Web resources.

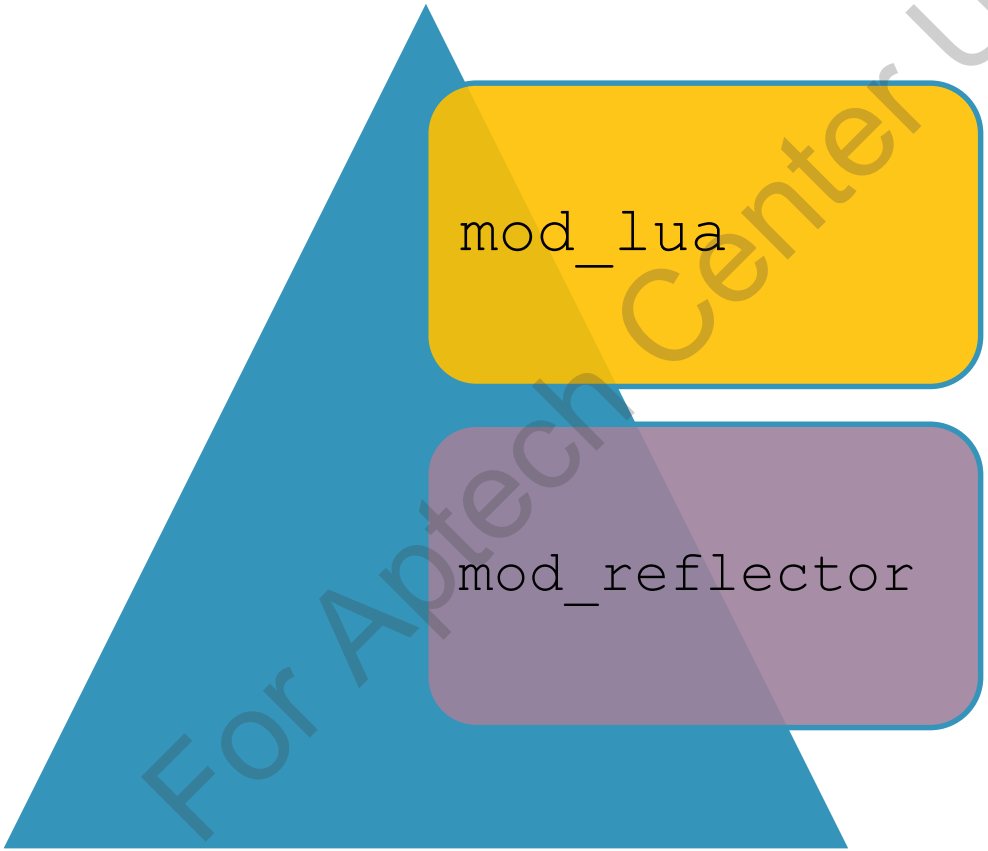
- Supports two modes of the FastCGI authorizer applications



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Introducing the Request Processing Modules

- Enhancements in the httpd request processing of Apache Web Server 2.4 are incorporated using the modules:



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Providing Support to Lua – mod_lua

- Provides support for rapid prototyping, scripting, and configuration because of the capability of managing the memory automatically by collecting garbage augmentation.

- Supports hooks to perform authentication, authorization, creating dynamic responses, mapping requests to files and access control of Apache Web Server 2.4.

- Example to load `mod_lua` and then handles requests of the .lua files:

```
LoadModule lua_moduleName modules/mod_lua.so

<Files "A*.lua">
    SetHandler lua-script
</Files>
```

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Returning the Request Services – mod_reflector

- Returns an HTTP request service as a response service by using a set of output filters.
- Uses the `ReflectorHeader` directive that returns an input header to a set of output headers.
- Syntax to use the `ReflectorHeader` directive:

```
ReflectorHeader in_header [out_header]
```

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Summary

- Apache Web Server 2.4 includes different modules that provide support for the server-side programming languages and authentication.
- mod_proxy_fcgi module is used to provide the FastCGI protocol backend services to mod_proxy.
- mod_proxy_scgi module provides the SCGI protocol backend services to mod_proxy and requires the mod_proxy and mod_proxy_scgi services (or module) should be available on the server to handle SCGI.
- You can use the proxy_html_module to provide the reverse proxy where an URL is rewritten to verify the URL is accessible outside the proxy as well as available publicly.

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Summary

- The mod_auth_form module works with the help of authentication and authorization modules, such as mod_authn_file and mod_authz_user.
- The mod_session module is used to provide support for the sessions, which can be created on the server or on the browser.
- The mod_authnz_fcgi module helps FastCGI authorizer applications to validate users using the Basic authentication method and to approve Apache Web resources.
- The mod_reflector module returns an HTTP request service as a response service by using a set of output filters.

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